Mathematical Induction

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Mathematical Induction: Concept

Definition

In general, **mathematical induction** is a technique used to prove that a statement $P(n)$ is true for all positive integers n, where $P(n)$ is a propositional function.

- It provides a systematic way to establish the truth of an infinite sequence of statements.

Structure of Inductive Proofs

A proof by mathematical induction has two main parts:

1. **Basis Step:** Show that $P(1)$ is true.
2. **Inductive Step:** Show that if $P(k)$ is true, then $P(k+1)$ is true for all positive integers k.

Mathematical Induction: Concept

Inductive Hypothesis

In the inductive step, we **assume** that $P(k)$ is true for an arbitrary positive integer k. This assumption is called the *inductive hypothesis*. Using it, we then prove that $P(k+1)$ must also be true.

$$P(1) \wedge (P(k) \rightarrow P(k+1)) \Rightarrow \forall n\, P(n)$$

Intuition

$$P(1) \rightarrow P(2)$$
$$P(2) \rightarrow P(3)$$
$$P(3) \rightarrow P(4)$$
$$\ldots\ldots$$

Example 1: Weak Induction

Statement: For all positive integers n,

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}.$$

Proof by Induction

Let $P(n)$ be the given proposition.

Basis Step ($n = 1$):

$$\text{LHS}(1) = 1, \qquad \text{RHS}(1) = \frac{1(1+1)}{2} = 1.$$

Thus $P(1)$ holds.

Inductive Step: Assume $P(k)$ holds (inductive hypothesis):

$$1 + 2 + \cdots + k = \frac{k(k+1)}{2}$$

We must show $P(k+1)$:

$$1 + 2 + \cdots + k + (k+1) = \frac{(k+1)(k+2)}{2}$$

Example 1: Weak Induction

From the IH $1 + 2 + \cdots + k = \dfrac{k(k+1)}{2}$, **add** $(k+1)$ **to both sides**:

$$1 + 2 + \cdots + k + (k+1) = \frac{k(k+1)}{2} + (k+1)$$

Simplify the RHS:

$$\frac{k(k+1)}{2} + (k+1) = \frac{k(k+1) + 2(k+1)}{2} = \frac{(k+1)(k+2)}{2} = \text{RHS}.$$

Therefore LHS = RHS, so $P(k+1)$ holds.

Conclusion

Since $P(1)$ holds and $P(k) \Rightarrow P(k+1)$, we conclude that $P(n)$ is true for all $n \in \mathbb{Z}^+$.

Example 2: Weak Induction

Statement: For all nonnegative integers n,

$$1 + 2 + 2^2 + \cdots + 2^n = 2^{n+1} - 1.$$

Proof by Induction

Let $P(n)$ be the given proposition.

Basis Step ($n = 0$):

$$\text{LHS}(0) = 1, \qquad \text{RHS}(0) = 2^{0+1} - 1 = 1.$$

Thus $P(0)$ holds.

Inductive Step: Assume $P(k)$ holds (inductive hypothesis):

$$1 + 2 + 2^2 + \cdots + 2^k = 2^{k+1} - 1.$$

We must show $P(k+1)$:

$$1 + 2 + 2^2 + \cdots + 2^k + 2^{k+1} = 2^{k+2} - 1.$$

Example 2: Weak Induction

From the IH $1 + 2 + \cdots + 2^k = 2^{k+1} - 1$, **add** 2^{k+1} **to both sides**:

$$1 + 2 + \cdots + 2^k + 2^{k+1} = (2^{k+1} - 1) + 2^{k+1}.$$

Simplify the RHS:

$$(2^{k+1} - 1) + 2^{k+1} = 2 \cdot 2^{k+1} - 1 = 2^{k+2} - 1 = \text{RHS}.$$

Therefore LHS = RHS, so $P(k+1)$ holds.

Conclusion

Since $P(0)$ holds and $P(k) \Rightarrow P(k+1)$, we conclude that $P(n)$ is true for all $n \in \mathbb{Z}_+ \cup \{0\}$

Example 3: Weak Induction

Statement: For all positive integers n,

$$1 \cdot 1! + 2 \cdot 2! + \cdots + n \cdot n! \;=\; (n+1)! - 1.$$

Proof by Induction

Let $P(n)$ be the given proposition.

Basis Step ($n = 1$):

$$\text{LHS}(1) = 1 \cdot 1! = 1, \qquad \text{RHS}(1) = (1+1)! - 1 = 2! - 1 = 1.$$

Thus $P(1)$ holds.

Inductive Step: Assume $P(k)$ holds (inductive hypothesis):

$$1 \cdot 1! + 2 \cdot 2! + \cdots + k \cdot k! \;=\; (k+1)! - 1.$$

We must show $P(k+1)$:

$$1 \cdot 1! + 2 \cdot 2! + \cdots + k \cdot k! + (k+1)(k+1)! \;=\; (k+2)! - 1.$$

Example 3: Weak Induction

From the IH $1 \cdot 1! + \cdots + k \cdot k! = (k+1)! - 1$, **add** $(k+1)(k+1)!$ **to both sides**:

$$1 \cdot 1! + \cdots + k \cdot k! + (k+1)(k+1)! = \big((k+1)! - 1\big) + (k+1)(k+1)!.$$

Simplify the RHS:

$$\begin{aligned}\big((k+1)! - 1\big) + (k+1)(k+1)! &= \big[1 + (k+1)\big](k+1)! - 1 \\ &= (k+2)(k+1)! - 1 = (k+2)! - 1 = \text{RHS}.\end{aligned}$$

Therefore LHS = RHS, so $P(k+1)$ holds.

Conclusion

Since $P(1)$ holds and $P(k) \Rightarrow P(k+1)$, we conclude that $P(n)$ is true for all $n \in \mathbb{Z}_+$

Strong Induction: Principle

Strong Induction

To prove $P(n)$ for all positive integers n:

- **Basis Step:** Verify $P(1)$ (or $P(a)$ if starting at $n = a$).
- **Inductive Step:** Show that

$$\big[P(1) \wedge P(2) \wedge \cdots \wedge P(k)\big] \;\Rightarrow\; P(k+1)$$

for all positive integers k.

Inductive Hypothesis (IH)

When proving the step for $k+1$, you may assume *all* $P(j)$ hold for $1 \leq j \leq k$ (not just $P(k)$). This makes strong induction more flexible than weak induction.

Strong vs. Weak Induction

- **Weak Induction:** Assume $P(k)$ to prove $P(k+1)$.
- **Strong Induction:** Assume $P(1), P(2), \ldots, P(k)$ to prove $P(k+1)$.

Same Logical Strength

Both methods are logically equivalent; strong induction is often easier when $P(k+1)$ naturally depends on *earlier* cases (e.g., $k-1$, $k-2$, $k-4$, ...).

Example (Strong Induction)

Statement: Every amount of postage of 12 cents or more can be formed using only $4¢$ and $5¢$ stamps.

Proof by Strong Induction

Let $P(n)$ be "n cents can be formed using $4¢$ and $5¢$ stamps," for $n \geq 12$.

Basis Step: Verify $P(12)$–$P(15)$:

$$12 = 3 \cdot 4, \qquad 13 = 2 \cdot 4 + 1 \cdot 5,$$
$$14 = 1 \cdot 4 + 2 \cdot 5, \qquad 15 = 3 \cdot 5.$$

Thus $P(n)$ holds for $n = 12, 13, 14, 15$.

Example (Strong Induction)

Inductive Step: Fix $k \geq 15$ and assume the **IH**: $P(j)$ is true for all $12 \leq j \leq k$. We must show $P(k+1)$.
Since $k \geq 15$, we have $k+1-4 = k-3 \geq 12$. By the IH, $P(k-3)$ is true: we can form $(k-3)$ cents with $4¢$ and $5¢$ stamps.

Add one $4¢$ stamp to that construction to obtain $(k-3)+4 = k+1$ cents. Hence $P(k+1)$ holds.

Conclusion

By strong induction, $P(n)$ is true for all $n \geq 12$: every amount of postage of 12 cents or more can be formed using only $4¢$ and $5¢$ stamps.